

Rigid-Body Kinematics and Dynamics

UAVs

MEAer - Autumn Semestre – 2022/2023

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Rigid Motion in 3-D - Kinematics

- **Goal:** derive the equations of motion for a 3-D rigid body

- Kinematics – linear and angular

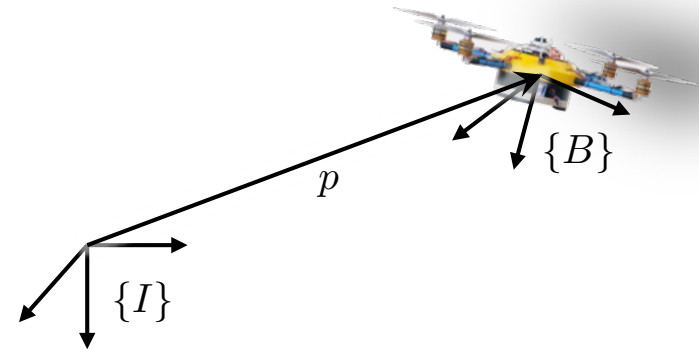
$$\dot{p} = Rv$$

$$\dot{R} = RS(\omega)$$

- Dynamics – linear and angular

$$m\dot{v} = -S(\omega)mv + f$$

$$J\dot{\omega} = -S(\omega)J\omega + n$$

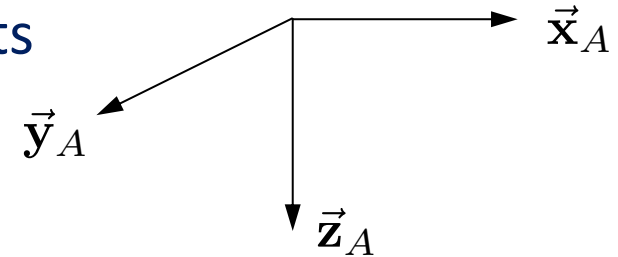


- *How do we get there ?*

Reference frames and configurations

- A **reference frame** {A} is defined by its

- Origin
- 3 orthonormal axes
(right-hand convention)



- The **configuration** of {B} w. r. t. {I} is defined by

with reference to

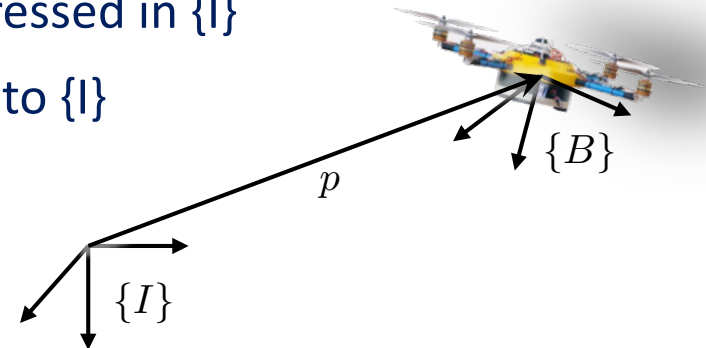
- The position (of the origin) of {B} expressed in {I}

$${}^I p_B \in \mathbb{R}^3 \quad \text{Column vector}$$

- The orientation (of the axes) of {B} expressed in {I}

> Rotation matrix representation from {B} to {I}

$${}^I_B R \in \mathbb{SO}^3$$



Notation: the superscript indicates that the vector is expressed in the specified reference frame

Reference frames and configurations

- Inertial reference frame (flat earth model)

 $\{I\}$

- Body-fixed reference frame

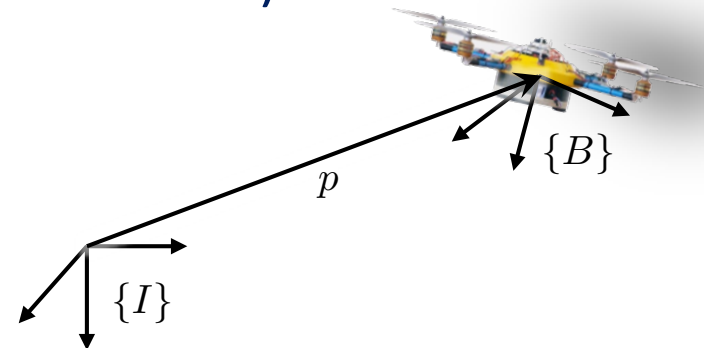
 $\{B\}$

- Configuration of $\{B\}$ w. r. t. $\{I\}$

$$({}^I p_B, {}^I_B R) \in \mathbb{R}^3 \times \mathbb{SO}(3)$$

${}^I p_B \in \mathbb{R}^3$ position of the origin of $\{B\}$ expressed in $\{I\}$

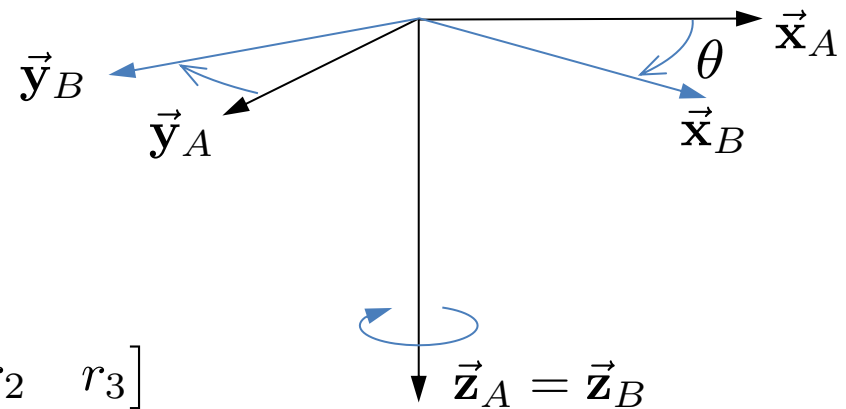
${}^I_B R \in \mathbb{SO}^3$ rotation matrix from $\{B\}$ to $\{I\}$



Rotation Matrices and reference frames

Rotation from frame $\{B\}$ to frame $\{A\}$

$${}^A_B R = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



In general,

$${}^A_B R = \begin{bmatrix} {}^A x_B & {}^A y_B & {}^A z_B \end{bmatrix} = \begin{bmatrix} r_1 & r_2 & r_3 \end{bmatrix}$$

Concatenates the coordinates of the principal axes of $\{B\}$ expressed in $\{A\}$

In this case,

$${}^A x_B = r_1 = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \\ 0 \end{bmatrix} \quad {}^A y_B = r_2 = \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \\ 0 \end{bmatrix} \quad {}^A z_B = r_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Rotation Matrices and reference frames

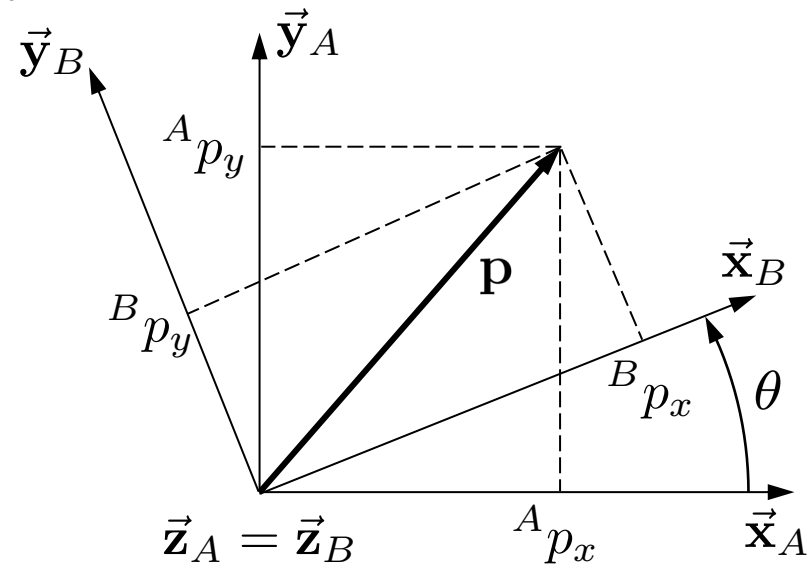
Rotation from frame $\{B\}$ to frame $\{A\}$

- Coordinates expressed in $\{B\}$ are transformed to coordinates expressed in $\{A\}$

$${}^A p = {}^A_B R {}^B p$$

$$\begin{bmatrix} {}^A p_x \\ {}^A p_y \\ {}^A p_z \end{bmatrix} = {}^A_B R \begin{bmatrix} {}^B p_x \\ {}^B p_y \\ {}^B p_z \end{bmatrix}$$

$$\begin{bmatrix} p \cos(\theta + \alpha) \\ p \sin(\theta + \alpha) \\ 0 \end{bmatrix} = {}^A_B R \begin{bmatrix} p \cos(\alpha) \\ p \sin(\alpha) \\ 0 \end{bmatrix}$$



Rotation about z axis

$${}^A_B R = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation Matrices – *Special Orthogonal Group*

Properties of $R = \begin{bmatrix} r_1 & r_2 & r_3 \end{bmatrix} \in \mathbb{R}^{3 \times 3}$

- Columns are mutually *orthonormal* $r_i^T r_i = 1, r_i^T r_j = 0, j \neq i$
 $\det(R) = r_1^T (r_2 \times r_3) = \pm 1$
- Right-handed coordinate system -> *special* $\det(R) = 1$
- *Group* (under matrix multiplication)
 - Closure: $R_1, R_2 \in \mathbb{SO}(3) \Rightarrow R_1 R_2 \in \mathbb{SO}(3)$
 - Identity: $R I_3 = I_3 R = R, \forall R \in \mathbb{SO}(3)$
 - Inverse: $R^{-1} = R^T$ inverse is unique and satisfies $R R^T = R^T R = I_3$
 - Associativity: $(R_1 R_2) R_3 = R_1 (R_2 R_3)$ **NOT commutative**
- *Special Orthogonal Group of order 3*
 $\mathbb{SO}(3) = \{R \in \mathbb{R}^{3 \times 3} : R R^T = R^T R = I_3, \det(R) = 1\}$

Rotation Matrices - Properties

Properties of $R = \begin{bmatrix} r_1 & r_2 & r_3 \end{bmatrix} \in \mathbb{R}^{3 \times 3}$

- Inverse

$${}^B_A R = ({}^A_B R)^{-1} = ({}^A_B R)^T$$

$${}^A p = {}^A_B R {}^B p \Leftrightarrow {}^B p = ({}^A_B R)^T {}^A p$$

- Composition rule

$${}^A_B R {}^B_C R = {}^A_C R$$

Rotation Matrices - Properties

Properties of $R = \begin{bmatrix} r_1 & r_2 & r_3 \end{bmatrix} \in \mathbb{R}^{3 \times 3}$

- Rotations are rigid body transformations (linear isometries)
 - R preserves distance between points:

$$\|R(p - q)\| = \|p - q\|$$

- R preserves angle between vectors:

- Outer product:

$$R(p \times q) = (Rp) \times (Rq)$$

- Inner product:

$${}^A\vec{u} \cdot {}^A\vec{v} = [{}^A u]^T [{}^A v] = [{}^B_A R^A u]^T [{}^B_A R^A v] = [{}^B u]^T [{}^B v] = {}^B\vec{u} \cdot {}^B\vec{v}$$

The skew-symmetric matrix

- Alternative way to express the cross-product

$$S(\mathbf{a})\mathbf{b} = \mathbf{a} \times \mathbf{b} = \begin{bmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{bmatrix}$$

$$S(\mathbf{a}) = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$

- Returns a skew-symmetric matrix

$$S := S(\mathbf{a})$$

$$S^T = -S$$

$$S^T + S = 0$$

- R preserves orientation** $RS(\mathbf{p}) = S(R\mathbf{p})R$ or $RS(\mathbf{p})R^T = S(R\mathbf{p})$

Rotation Representations

Exponential map

Euler's rotation theorem

- Any rotation can be obtained by rotating an angle $\theta \in [0, 2\pi)$ about an axis $\mathbf{n} \in \mathbb{R}^3$, $\|\mathbf{n}\| = 1$

$$R(\theta, \mathbf{n}) = e^{\theta S(\mathbf{n})} = I_3 + \theta S(\mathbf{n}) + \frac{\theta^2}{2} S(\mathbf{n})^2 + \frac{\theta^3}{3!} S(\mathbf{n})^3 + \dots$$

- Rodrigues' formula

$$R(\theta, \mathbf{n}) = I_3 + \sin \theta S(\mathbf{n}) + (1 - \cos \theta) S(\mathbf{n})^2$$

$$R(\theta, \mathbf{n})\mathbf{p} = \mathbf{n}\mathbf{n}^T \mathbf{p} + \sin \theta S(\mathbf{n})\mathbf{p} - \cos \theta S(\mathbf{n})^2 \mathbf{p}$$

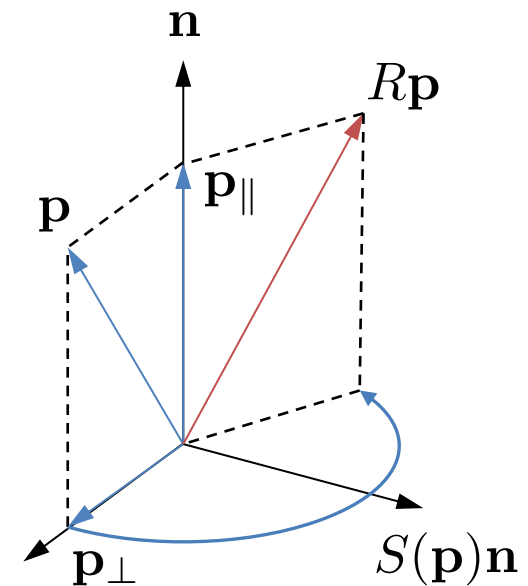
Rotation Representations

Exponential map and Rodrigues' formula

$$\begin{aligned} R(\theta, \mathbf{n}) &= I_3 + \sin \theta S(\mathbf{n}) + (1 - \cos \theta) S(\mathbf{n})^2 \\ &= \mathbf{n}\mathbf{n}^T + \sin \theta S(\mathbf{n}) - \cos \theta S(\mathbf{n})^2 \end{aligned}$$

$$\begin{aligned} \mathbf{p} &= \mathbf{p}_{\parallel} + \mathbf{p}_{\perp} \\ &= \mathbf{n}\mathbf{n}^T \mathbf{p} - S(\mathbf{n})^2 \mathbf{p} \end{aligned}$$

$$R\mathbf{p} = \mathbf{p}_{\parallel} + \sin \theta S(\mathbf{n})\mathbf{p} + \cos \theta \mathbf{p}_{\perp}$$



Unit quaternions are an alternative representation similar to the definition of vector and angle

$$\mathbf{q} = \left(\cos \left(\frac{\theta}{2} \right), \sin \left(\frac{\theta}{2} \right) \mathbf{n} \right)$$

Rotation Representations

Euler angles

- Any rotation can be decomposed into three elementary rotations, e.g. Z-Y-Z, Y-Z-X, Z-Y-X.
- Z-Y-X parametrization, also known as, yaw ψ , pitch θ , roll ϕ Euler angles

$$\lambda = [\phi \quad \theta \quad \psi]^T \in \mathbb{R}^3$$

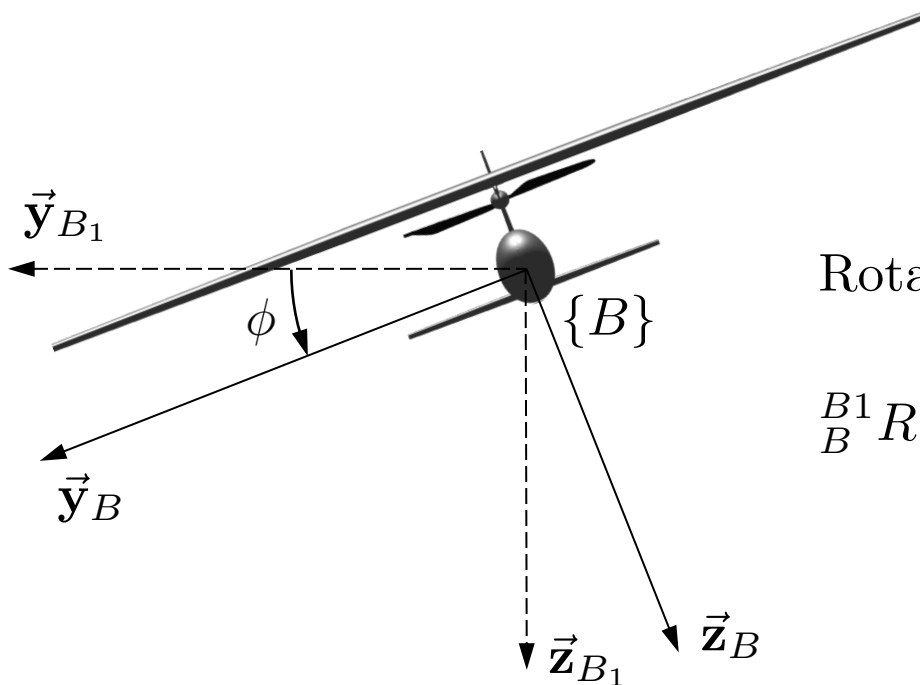
$${}^I_B R = R(\lambda) = R_z(\psi)R_y(\theta)R_x(\phi)$$

$$= \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix}$$

- 3-dimensional parametrization means that singularities are always present

Z-Y-X Euler angles – Intermediate frames

Rotation from $\{B\}$ to $\{B_1\}$



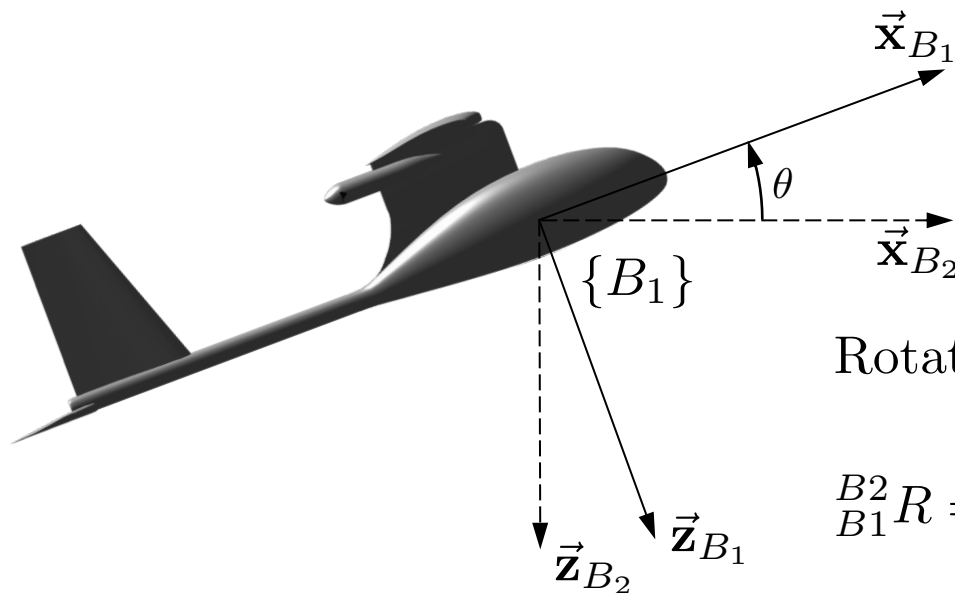
Rotation about x-axis of body frame (roll)

$${}^{B_1}_B R = R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix}$$

Beard & McLain, "Small Unmanned Aircraft," Princeton University Press, 2012

Z-Y-X Euler angles – Intermediate frames

Rotation from $\{B_1\}$ to $\{B_2\}$



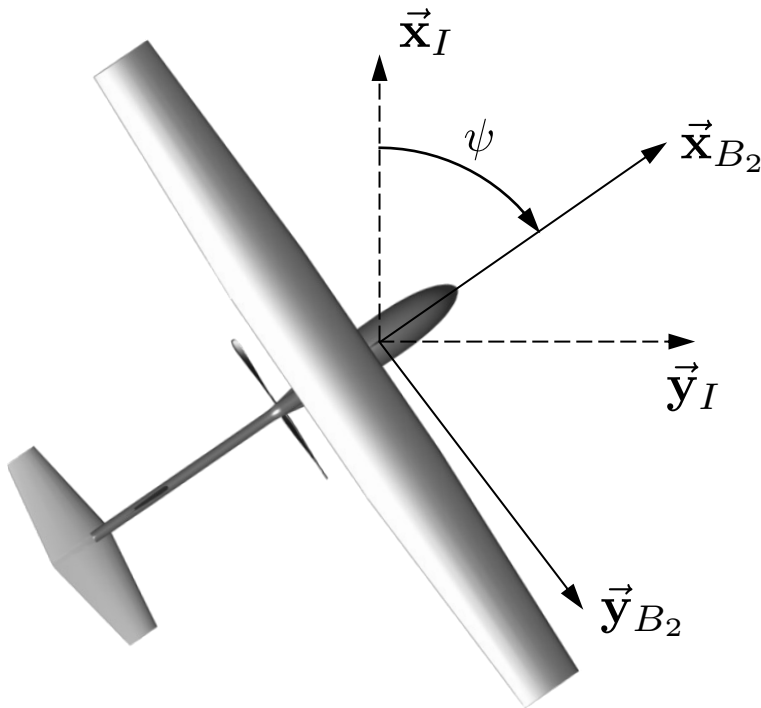
Rotation about y -axis of frame $\{B_1\}$ (pitch)

$${}^{B_2}_{B_1}R = R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

Z-Y-X Euler angles – Intermediate frames

Rotation from $\{B_2\}$ to $\{B_3\}$

$\{B_3\}$ and $\{I\}$: same orientation, different origins



Rotation about z-axis of frame $\{B_2\}$ (yaw)

$${}^I_{B_2}R = R_z(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

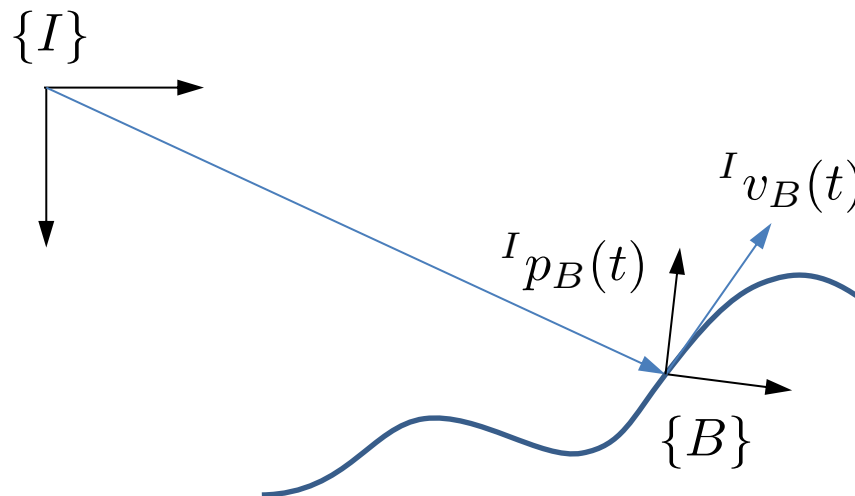
Beard & McLain, "Small Unmanned Aircraft," Princeton University Press, 2012

Rigid Body Kinematics – Linear Motion

- Linear motion: position evolves as a function of time

$${}^I\dot{p}_B(t) = {}^Iv_B(t)$$

${}^Iv_B(t)$ linear velocity expressed in inertial frame
 tangent to the curve at ${}^Ip_B(t)$



Rigid Body Kinematics – Angular Motion

- Angular motion: orientation evolves as a function of time

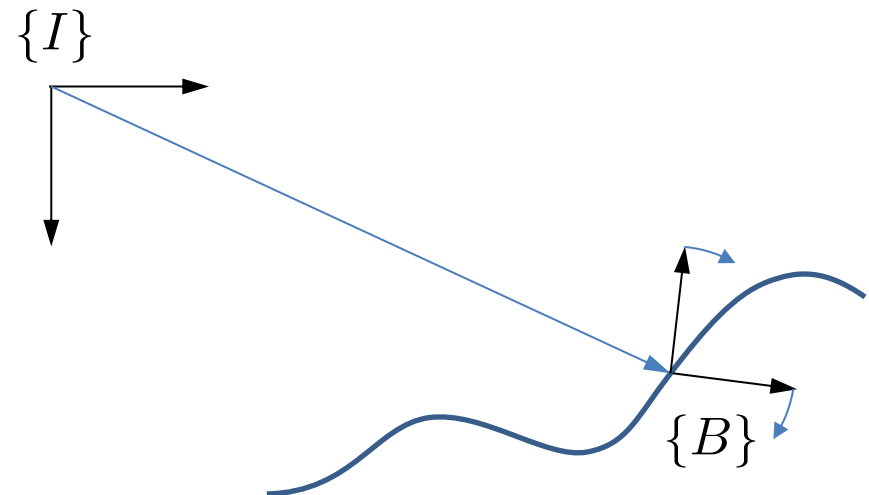
$${}^I_B \dot{R} = S({}^I \omega_B) {}^I_B R$$

${}^I \omega_B(t)$ angular velocity expressed in inertial frame

Recall that

$$S(a)b = a \times b$$

$$S(a) = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$



Rigid Body Kinematics – Angular Motion

- Angular motion

$${}^I_B \dot{R} = S({}^I \omega_B) {}^I_B R$$

Why is that so?

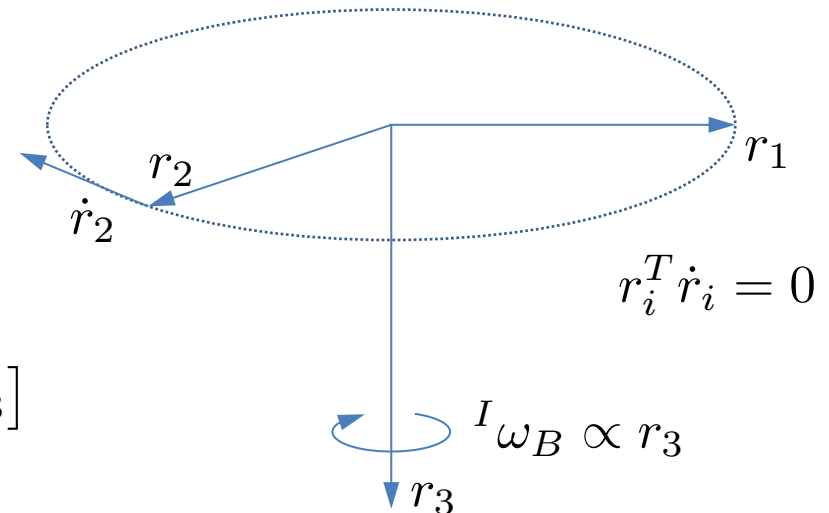
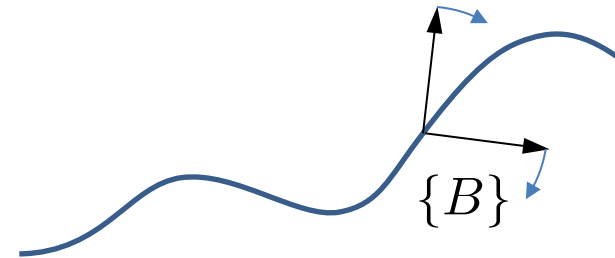
$${}^I_B R = R = [r_1 \quad r_2 \quad r_3] \in \mathbb{SO}(3)$$

$$r_i^T r_i = 1$$

$$r_i^T \dot{r}_i = 0$$

$$\dot{r}_i = {}^I \omega_B \times r_i$$

$$[\dot{r}_1 \quad \dot{r}_2 \quad \dot{r}_3] = S({}^I \omega_B) [r_1 \quad r_2 \quad r_3]$$



Rigid Body Kinematics – Angular Motion

- Angular motion

$${}^I_B \dot{R} = S({}^I \omega_B) {}^I_B R$$

and
$${}^I_B \dot{R} = {}^I_B R S({}^B_I R {}^I \omega_B)$$

Using “lighter” notation

$$R := {}^I_B R, \quad \omega := {}^B_I R {}^I \omega_B$$

$$\dot{R} = S({}^I \omega_B) R$$

and

$$\dot{R} = R S(\omega)$$

Expressed in inertial frame

$$\begin{aligned} R R^T &= I \\ \dot{R} R^T &= -R \dot{R}^T = S({}^I \omega_B) \end{aligned}$$

Expressed in body frame

$$\begin{aligned} R^T R &= I \\ R^T \dot{R} &= -\dot{R}^T R = S(\omega) \end{aligned}$$

Rigid Body Kinematics

- With velocities *expressed in inertial frame*

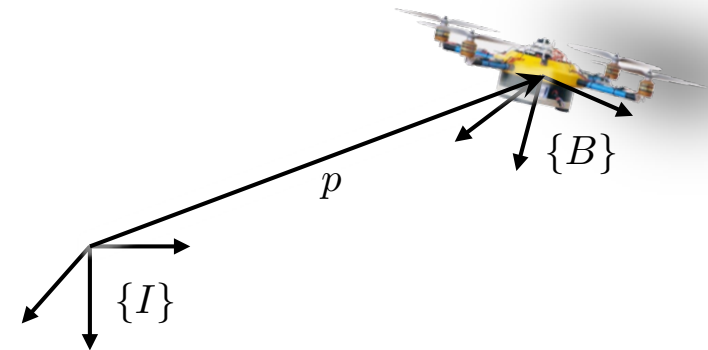
$$\dot{p} = {}^I v_B$$

$$\dot{R} = S({}^I \omega_B) R$$

- With velocities *in body frame*

$$\dot{p} = R v$$

$$\dot{R} = R S(\omega)$$



$${}^B p := {}^B_I R p$$

- With position also expressed in body frame

$${}^B \dot{p} := -S(\omega) {}^B p + v$$

Rigid Body Kinematics + Exercise

- Circular motion: given

$$p(t) = {}^I p_B(t) = r \begin{bmatrix} \cos \psi(t) \\ \sin \psi(t) \\ 0 \end{bmatrix}$$

$$R(t) = {}^I_B R(t) = R_z(\psi(t))$$

compute

$${}^B p = {}^B_I R p$$

$$v = {}^B_I R \dot{p}$$

$$\omega = {}^B_I R^I \omega_B$$

Rigid Body Kinematics - Exercise

- Circular motion: result...

$$p = \begin{bmatrix} rc_\psi \\ rs_\psi \\ 0 \end{bmatrix} = {}^I_B R p_0 = R(\psi) p_0 = \begin{bmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r \\ 0 \\ 0 \end{bmatrix}$$

$${}^B p = {}^B_I R p = R^T R p_0 = p_0$$

$$\dot{p} = \begin{bmatrix} -r\dot{\psi}s_\psi \\ r\dot{\psi}c_\psi \\ 0 \end{bmatrix} = \begin{bmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -\dot{\psi} & 0 \\ \dot{\psi} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} r \\ 0 \\ 0 \end{bmatrix} = RS(\omega) p_0$$

$$v = R^T \dot{p} = S(\omega) p_0 = \begin{bmatrix} 0 \\ r\dot{\psi} \\ 0 \end{bmatrix}$$

$${}^I \omega_B = S^{-1}(\dot{R}R^T) = \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} = \omega = R\omega$$

Rigid Body Kinematics

- Attitude kinematics (angular velocity in body frame)

$$\dot{R} = RS(\omega) \quad \text{Rotation matrix} \quad R \in \mathbb{SO}(3)$$

Or

$$\dot{\lambda} = Q(\lambda)\omega \quad \text{Z-Y-X Euler angles} \quad \lambda = [\phi \quad \theta \quad \psi]^T \in \mathbb{R}^3$$

$$Q(\lambda) = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \frac{\sin \phi}{\cos \theta} & \frac{\cos \phi}{\cos \theta} \end{bmatrix}$$

Rigid Body Kinematics – Exercise

- Given $\lambda = [\phi \quad \theta \quad \psi]^T \in \mathbb{R}^3$

$$R(\lambda) = R_z(\psi)R_y(\theta)R_x(\phi)$$

derive $\dot{\lambda} = Q(\lambda)\omega \quad Q(\lambda) = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \frac{\sin \phi}{\cos \theta} & \frac{\cos \phi}{\cos \theta} \end{bmatrix}$

$$\omega = Q^{-1}(\lambda) \dot{\lambda} = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & s_\phi c_\theta \\ 0 & -s_\phi & c_\phi c_\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

Intermediate step: derive

$$\omega = R_x^T(\phi)R_y^T(\theta)\dot{\psi}e_3 + R_x^T(\phi)\dot{\theta}e_2 + \dot{\phi}e_1 \quad e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

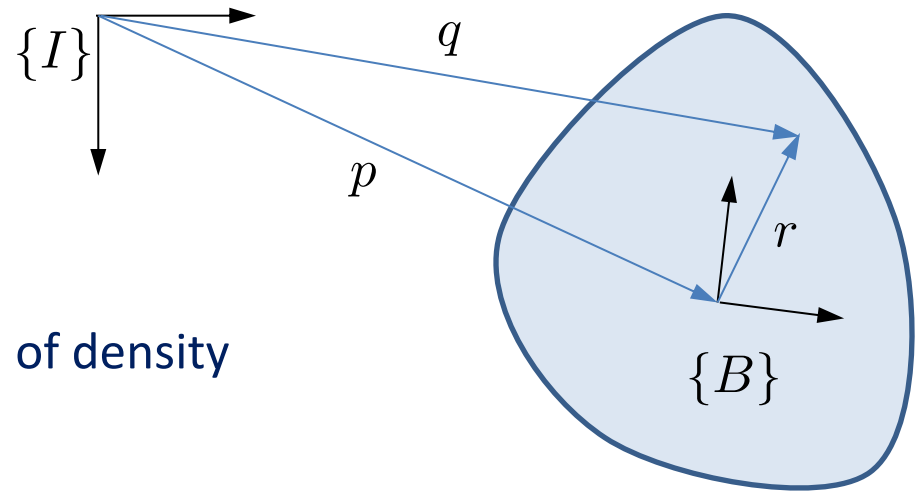
Note that Q is not a rotation!

Rigid Body Dynamics

- Particles in the rigid body

$${}^I q = {}^I p_B + {}^I_B R {}^B r$$

$$q = p + R r$$



- Mass

$$m = \int \rho(r) dV \quad \text{Volume integral of density}$$

- Center of mass

$$\bar{q} = \frac{1}{m} \int q(r) \rho(r) dV \quad \text{Weighted average of the density}$$

If origin of {B} coincides with center of mass

$$p = \bar{q} \quad \Rightarrow \quad \int r \rho(r) dV = 0$$

Rigid Body Dynamics

- Particles in the rigid body

$${}^I q = {}^I p_B + {}^I_B R {}^B r$$

$$q = p + R r$$

- Velocity of particles

$$\dot{q} = \dot{p} + R S(\omega) r$$

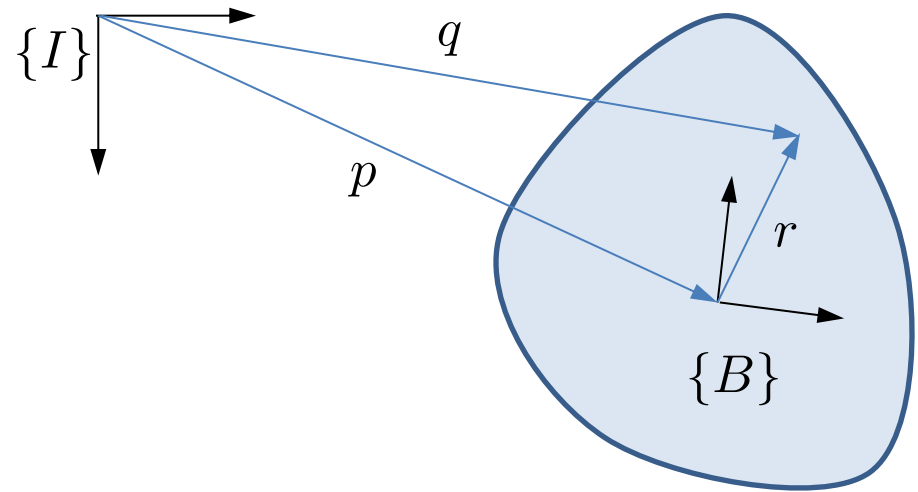
- Linear momentum

$$\int \dot{q}(r) \rho(r) dV = m \dot{p}$$

- Angular momentum relative to center of mass

$$\int \rho(r) (q(r) - \bar{q}) \times \dot{q}(r) dV = -R \int \rho(r) S(r)^2 dV \omega = R J \omega$$

Tensor of inertia $J = - \int \rho(r) S(r)^2 dV$



Rigid Body Dynamics – Tensor of Inertia

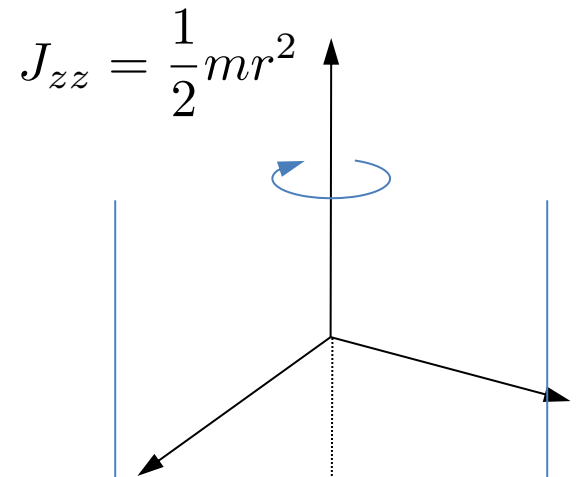
- Tensor of Inertia

$$J = - \int \rho(r) S(r)^2 dV$$

$$= \int \rho(r) (r^T r I_3 - r r^T) dV$$

$$= \int \rho(r) \begin{bmatrix} r_y^2 + r_z^2 & -r_x r_y & -r_x r_z \\ -r_x r_y & r_x^2 + r_z^2 & -r_y r_z \\ -r_x r_z & -r_y r_z & r_x^2 + r_y^2 \end{bmatrix} dV$$

$$= \begin{bmatrix} J_{xx} & J_{xy} & J_{xz} \\ J_{xy} & J_{yy} & J_{yz} \\ J_{xz} & J_{yz} & J_{zz} \end{bmatrix}$$



$$J_{xx} = J_{yy} = \frac{1}{12} m (3r^2 + h^2)$$

$$J_{xy} = J_{xz} = J_{yz} = 0$$

$$u_i = e_i$$

Principal axes are given by the eigenvalues of J , $Ju_i = \alpha_i u_i$

If the body rotates about a principal axis, angular momentum keeps the direction of rotation.

Rigid Body Dynamics

Newton-Euler formulation

- Newton's 2nd law
 - Conservation of momentum (holds in inertial frames)

- Translational motion

- Linear momentum

$$m\dot{p}$$

- Conservation of linear momentum

$$\frac{d}{dt}(m\dot{p}) = m\ddot{p} = {}^I f \quad \text{Applied forces, expressed in } \{I\}$$

- Transformation to body frame

$$\frac{d}{dt}(mRv) = {}^I f$$

$$mR\dot{v} + mRS(\omega)v = R {}^I f$$

$$\boxed{m\dot{v} + mS(\omega)v = f,} \quad f := R^T {}^I f$$

Rigid Body Dynamics

Newton-Euler formulation

- Newton's 2nd law
 - Conservation of momentum (holds in inertial frames)

- Rotational motion

- Angular momentum

$$R J \omega$$

- Conservation of angular momentum

$$\frac{d}{dt}(R J \omega) = R J \dot{\omega} + R S(\omega) J \omega = {}^I \tau \quad \text{Applied torques, expressed in } \{I\}$$

- Transformation to body frame

$$\boxed{J \dot{\omega} + S(\omega) J \omega = \tau,} \quad \tau = R^T {}^I \tau$$

Rigid Body Equations of Motion

- Goal achieved:
derive the equations of motion for a 3-D rigid body

- Kinematics

$$\dot{p} = Rv$$

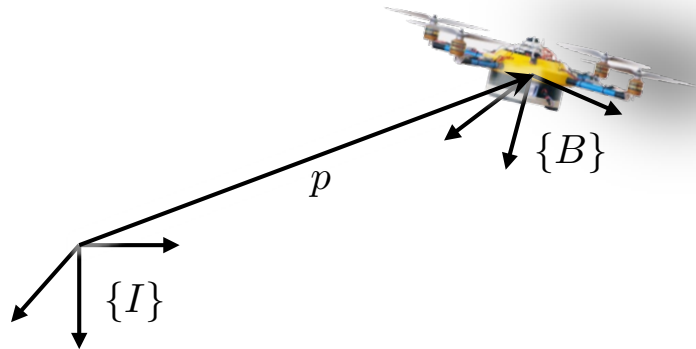
$$\dot{R} = RS(\omega)$$

- Dynamics **in body frame**

$$m\dot{v} = -S(\omega)mv + f$$

$$J\dot{\omega} = -S(\omega)J\omega + \tau$$

- State : $X = (p, R, v, \omega)$



Input : $U = (f, \tau)$